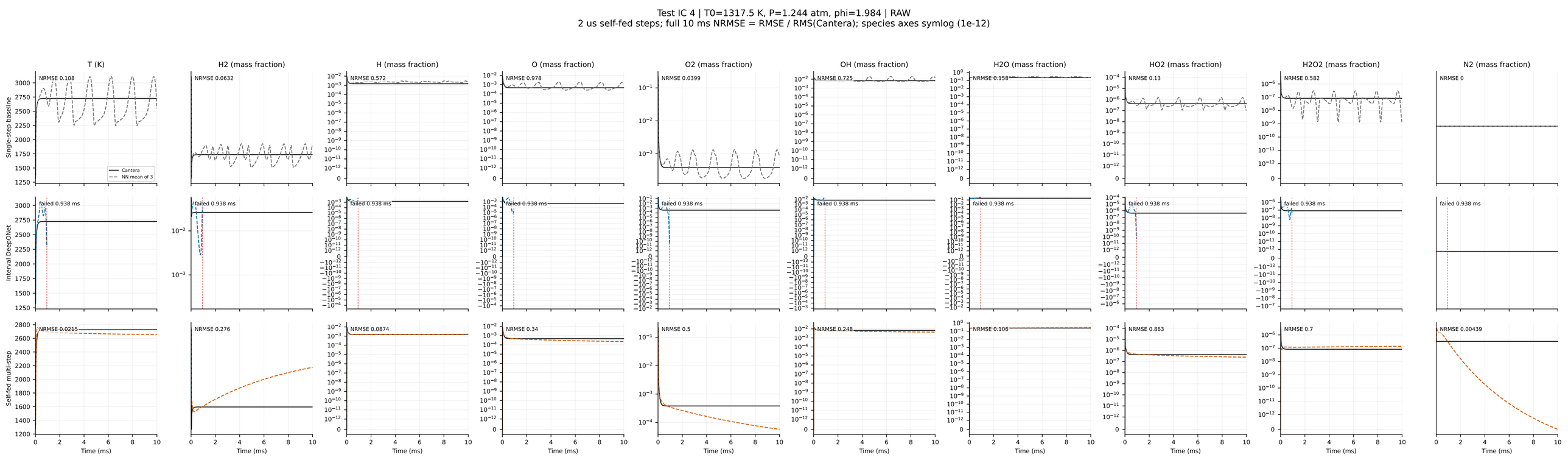
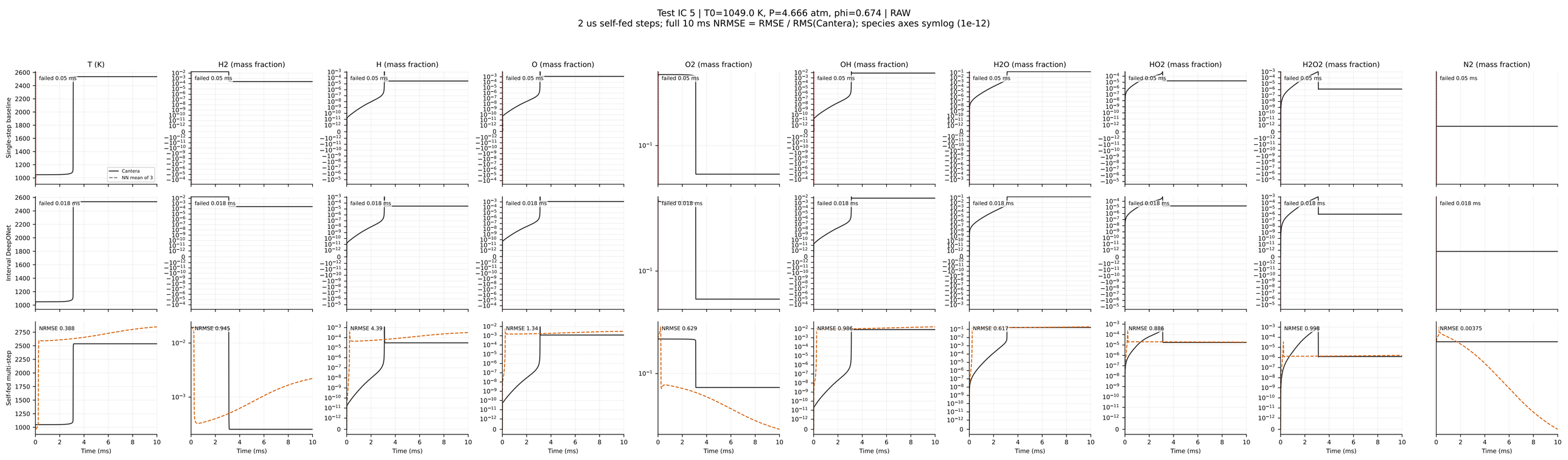
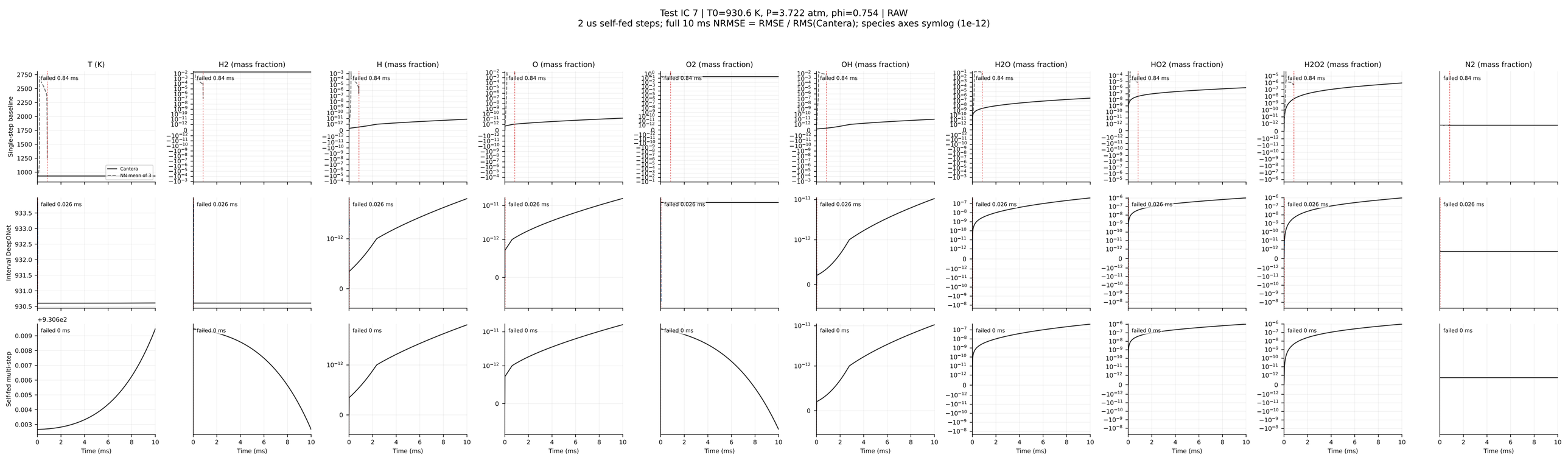
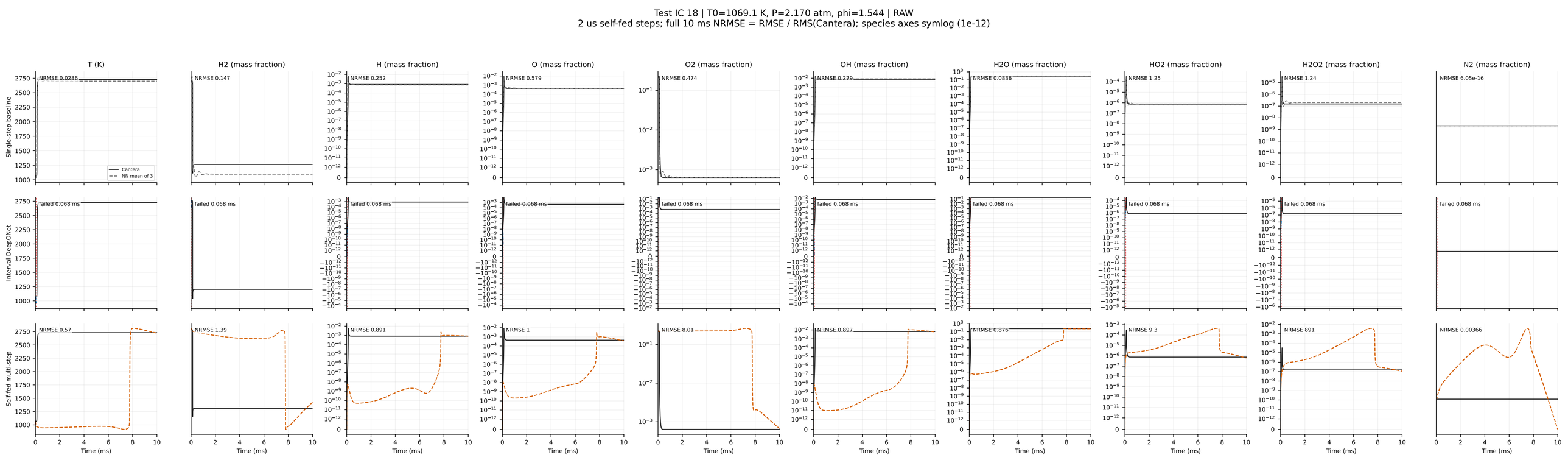
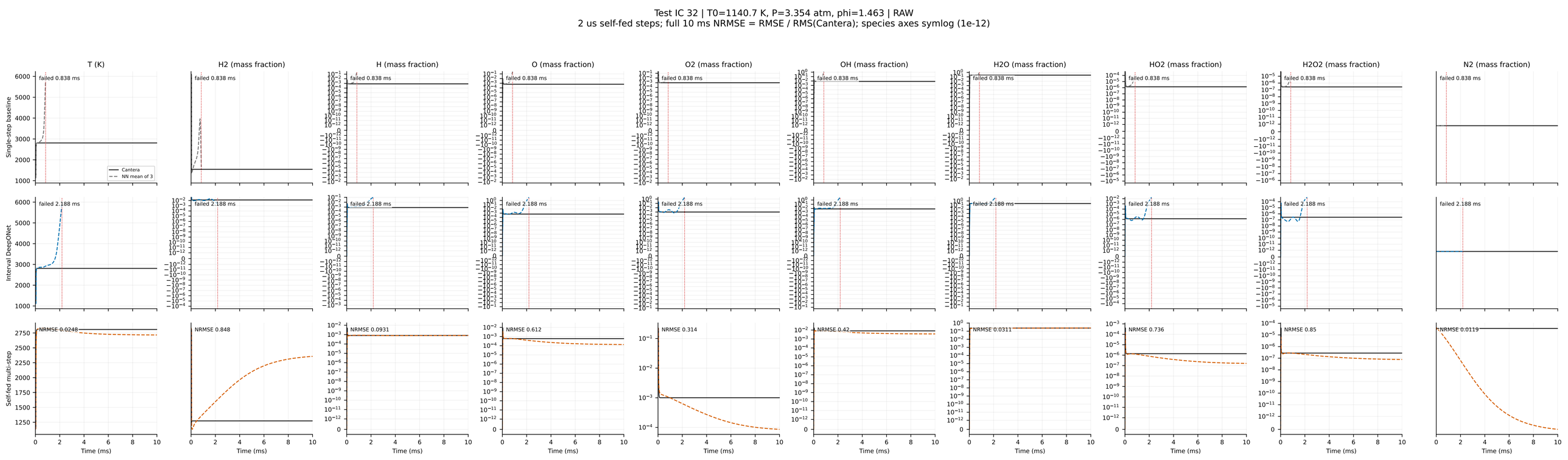
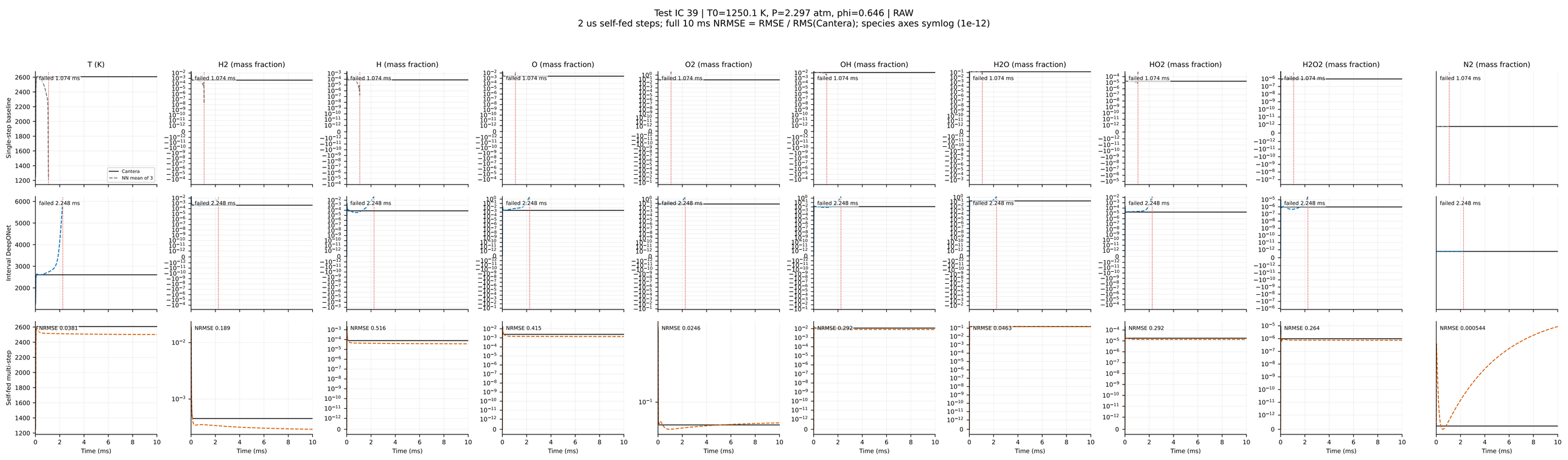

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.108
NRMSE 0.0632
NRMSE 0.572
NRMSE 0.978
NRMSE 0.0399
NRMSE 0.725
NRMSE 0.158
NRMSE 0.13
NRMSE 0.582
NRMSE 0
Cantera
NN mean of 3
failed 0.938 ms
NRMSE 0.0215
NRMSE 0.276
NRMSE 0.0874
NRMSE 0.34
NRMSE 0.5
NRMSE 0.248
NRMSE 0.106
NRMSE 0.863
NRMSE 0.7
NRMSE 0.00439
Time (ms)

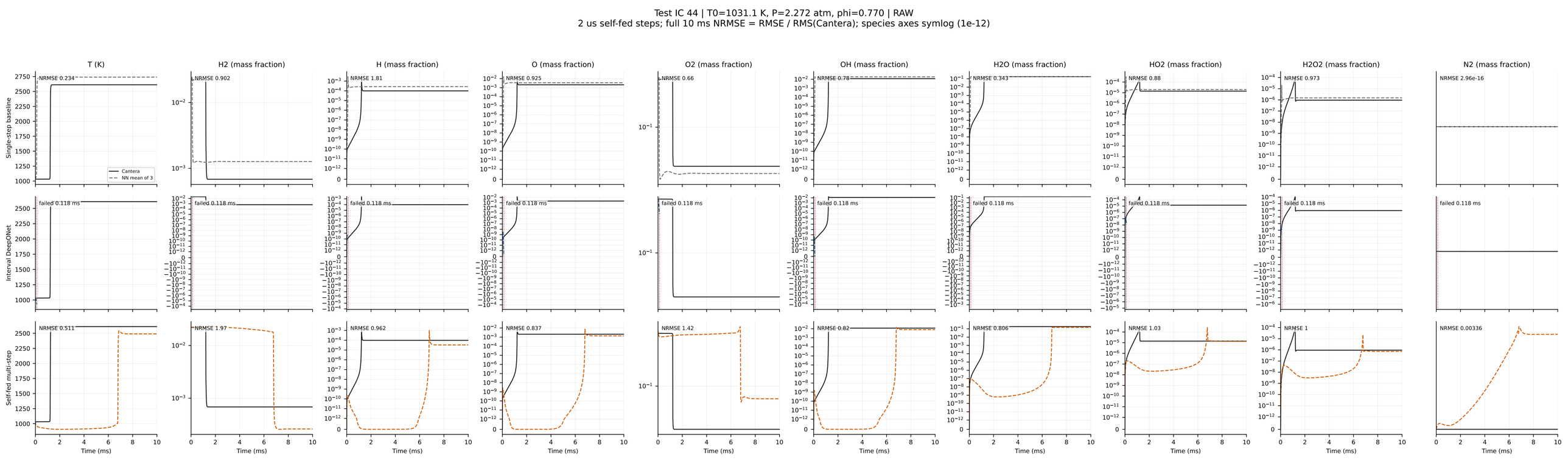

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.05 ms
failed 0.018 ms
Cantera
NN mean of 3
NRMSE 0.388
NRMSE 0.945
NRMSE 4.39
NRMSE 1.34
NRMSE 0.629
NRMSE 0.986
NRMSE 0.617
NRMSE 0.886
NRMSE 0.998
NRMSE 0.00375
Time (ms)

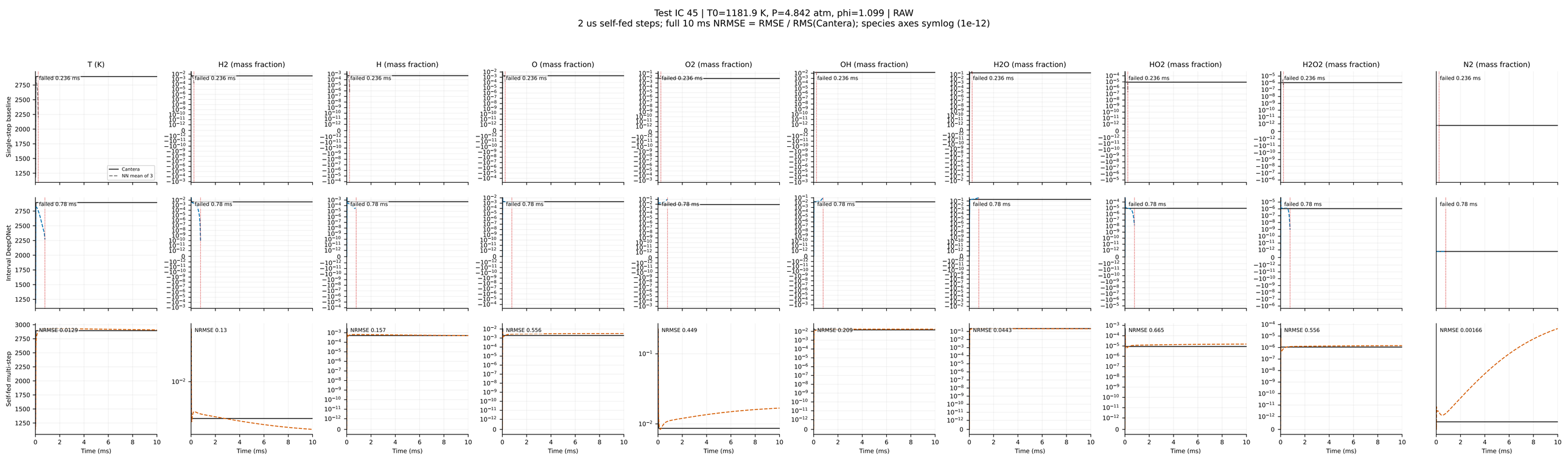

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.84 ms
failed 0.026 ms
failed 0 ms
Cantera
NN mean of 3
+9.306e2
Time (ms)

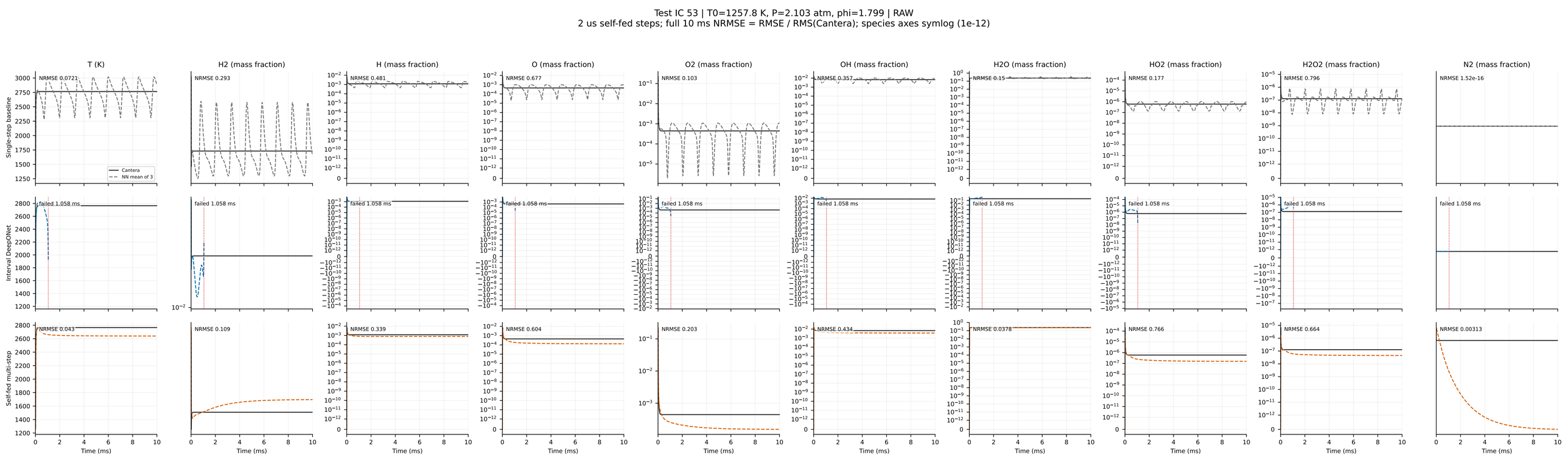

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0286
NRMSE 0.147
NRMSE 0.252
NRMSE 0.579
NRMSE 0.474
NRMSE 0.279
NRMSE 0.0836
NRMSE 1.25
NRMSE 1.24
NRMSE 6.05e-16
Cantera
NN mean of 3
failed 0.068 ms
NRMSE 0.57
NRMSE 1.39
NRMSE 0.891
NRMSE 1
NRMSE 8.01
NRMSE 0.897
NRMSE 0.876
NRMSE 9.3
NRMSE 891
NRMSE 0.00366
Time (ms)

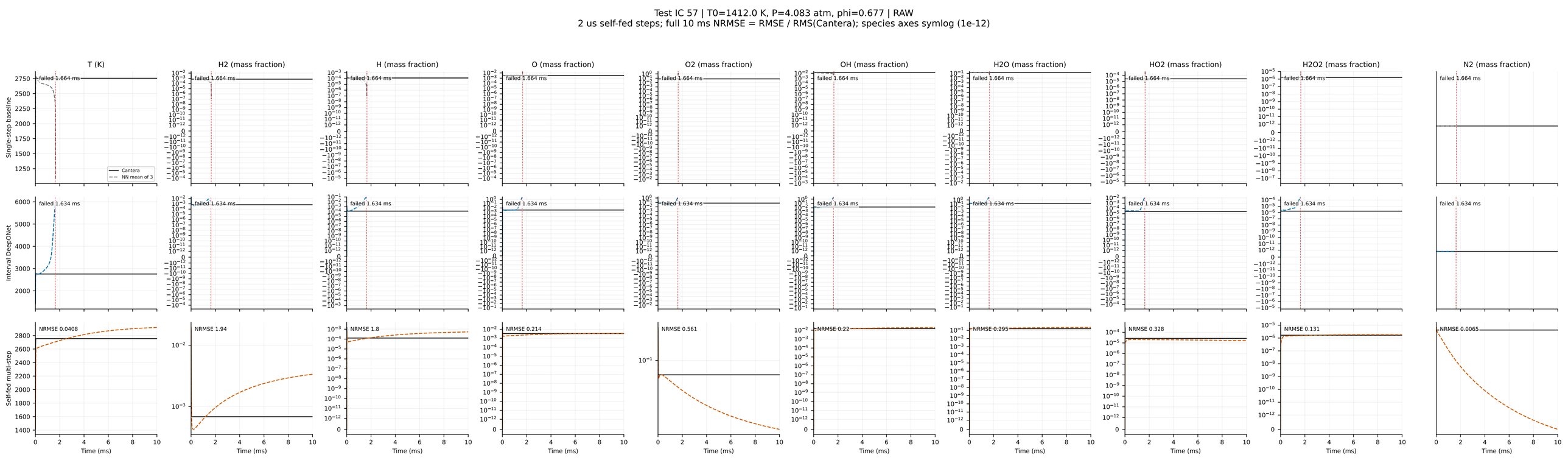

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 0.838 ms
failed 2.188 ms
NRMSE 0.0248
NRMSE 0.848
NRMSE 0.0931
NRMSE 0.612
NRMSE 0.314
NRMSE 0.42
NRMSE 0.0311
NRMSE 0.736
NRMSE 0.85
NRMSE 0.0119
Time (ms)

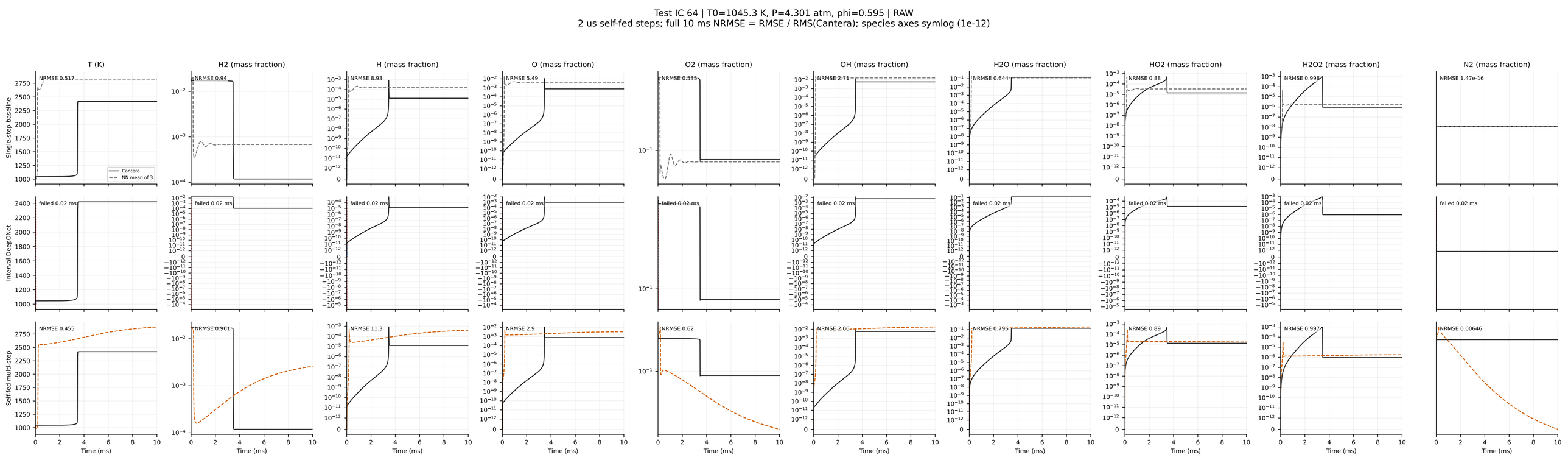

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 1.074 ms
failed 2.248 ms
NRMSE 0.0381
NRMSE 0.189
NRMSE 0.516
NRMSE 0.415
NRMSE 0.0246
NRMSE 0.292
NRMSE 0.0463
NRMSE 0.292
NRMSE 0.264
NRMSE 0.000544
Time (ms)

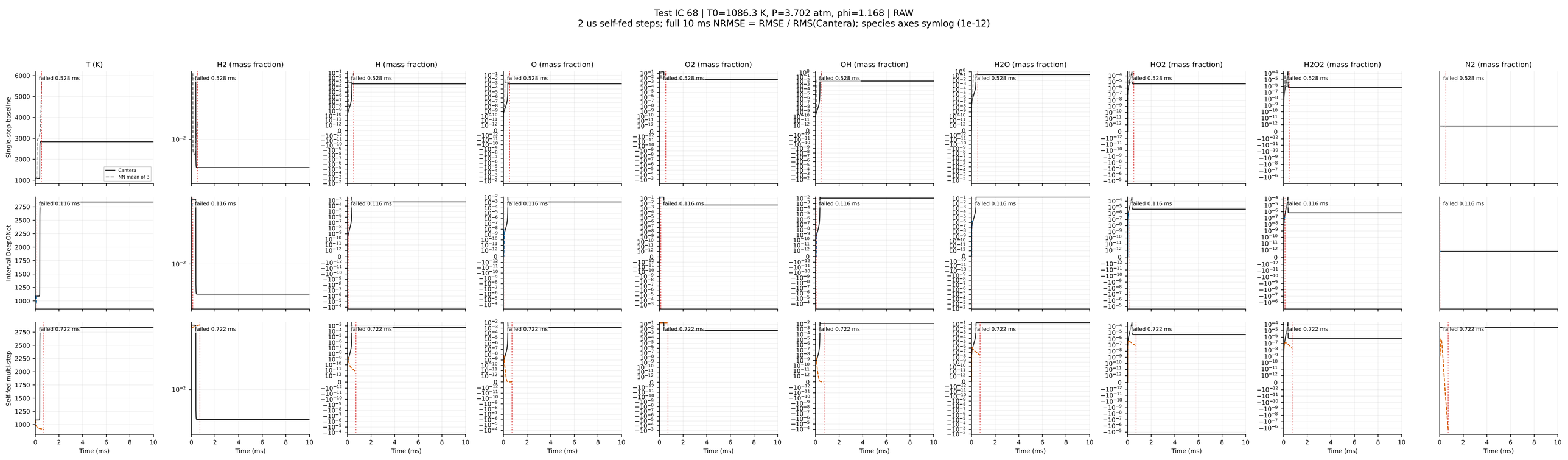

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.234
NRMSE 0.902
NRMSE 1.81
NRMSE 0.925
NRMSE 0.66
NRMSE 0.78
NRMSE 0.343
NRMSE 0.88
NRMSE 0.973
NRMSE 2.96e-16
failed 0.118 ms
NRMSE 0.511
NRMSE 1.97
NRMSE 0.962
NRMSE 0.837
NRMSE 1.42
NRMSE 0.82
NRMSE 0.806
NRMSE 1.03
NRMSE 1
NRMSE 0.00336
Time (ms)

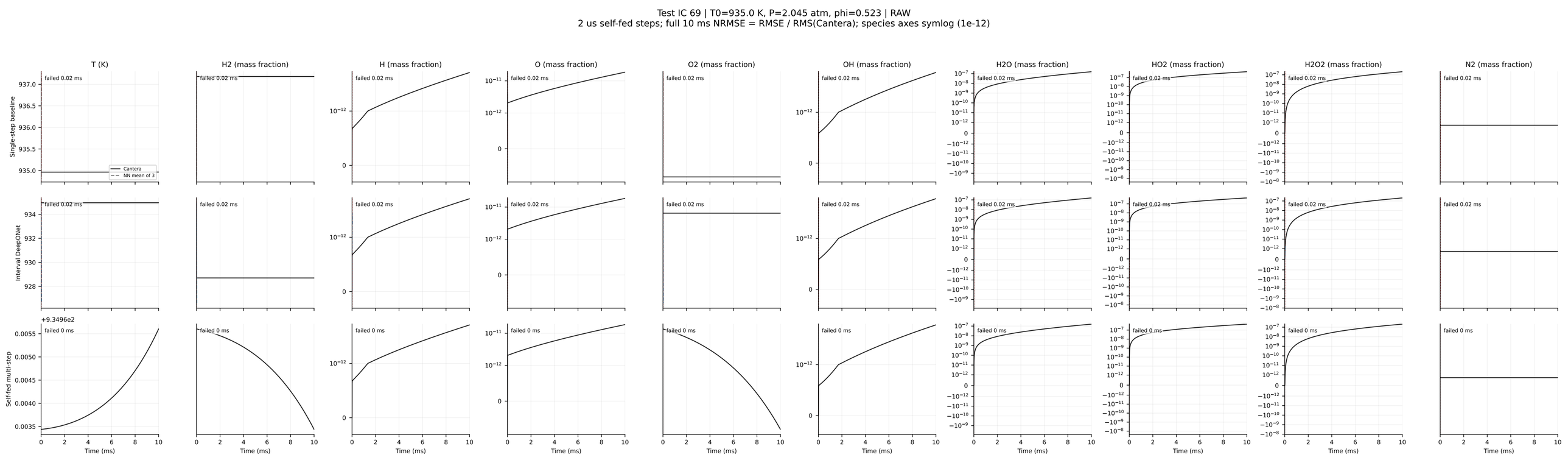

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.236 ms
failed 0.78 ms
Cantera
NN mean of 3
NRMSE 0.0129
NRMSE 0.13
NRMSE 0.157
NRMSE 0.556
NRMSE 0.449
NRMSE 0.209
NRMSE 0.0443
NRMSE 0.665
NRMSE 0.556
NRMSE 0.00166
Time (ms)

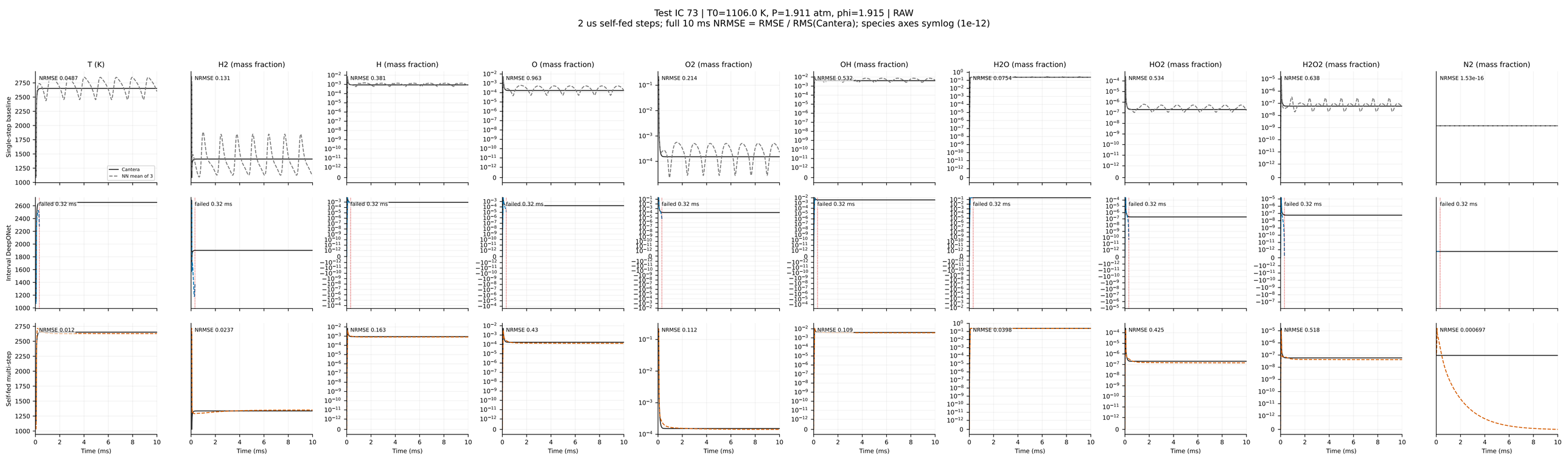

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0721
NRMSE 0.293
NRMSE 0.481
NRMSE 0.677
NRMSE 0.103
NRMSE 0.357
NRMSE 0.15
NRMSE 0.177
NRMSE 0.796
NRMSE 1.52e-16
Cantera
NN mean of 3
failed 1.058 ms
NRMSE 0.043
NRMSE 0.109
NRMSE 0.339
NRMSE 0.604
NRMSE 0.203
NRMSE 0.434
NRMSE 0.0378
NRMSE 0.766
NRMSE 0.664
NRMSE 0.00313
Time (ms)

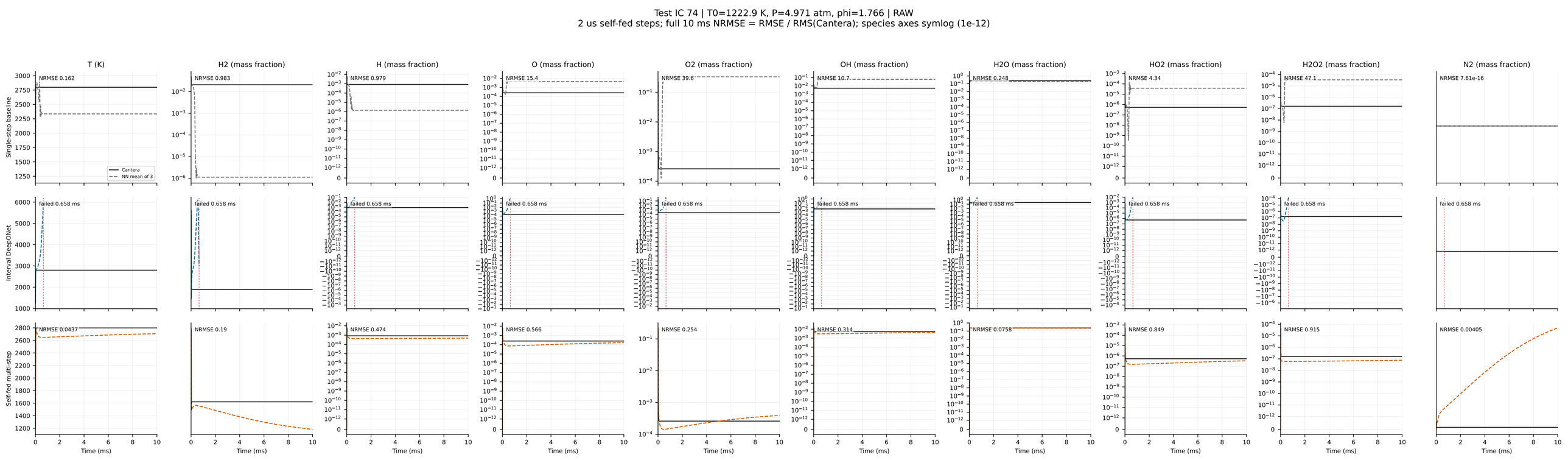

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 1.664 ms
failed 1.634 ms
Cantera
NN mean of 3
NRMSE 0.0408
NRMSE 1.94
NRMSE 1.8
NRMSE 0.214
NRMSE 0.561
NRMSE 0.22
NRMSE 0.295
NRMSE 0.328
NRMSE 0.131
NRMSE 0.0065
Time (ms)

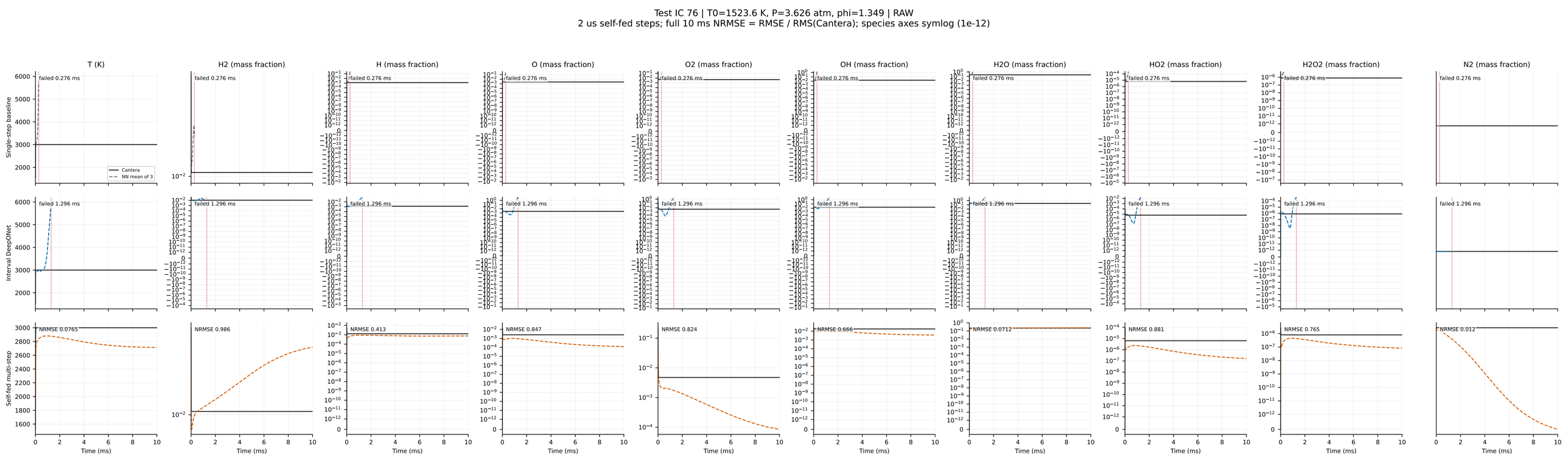

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.517
NRMSE 0.94
NRMSE 8.93
NRMSE 5.49
NRMSE 0.535
NRMSE 2.71
NRMSE 0.644
NRMSE 0.88
NRMSE 0.996
NRMSE 1.47e-16
failed 0.02 ms
NRMSE 0.455
NRMSE 0.961
NRMSE 11.3
NRMSE 2.9
NRMSE 0.62
NRMSE 2.06
NRMSE 0.796
NRMSE 0.89
NRMSE 0.997
NRMSE 0.00646
Time (ms)

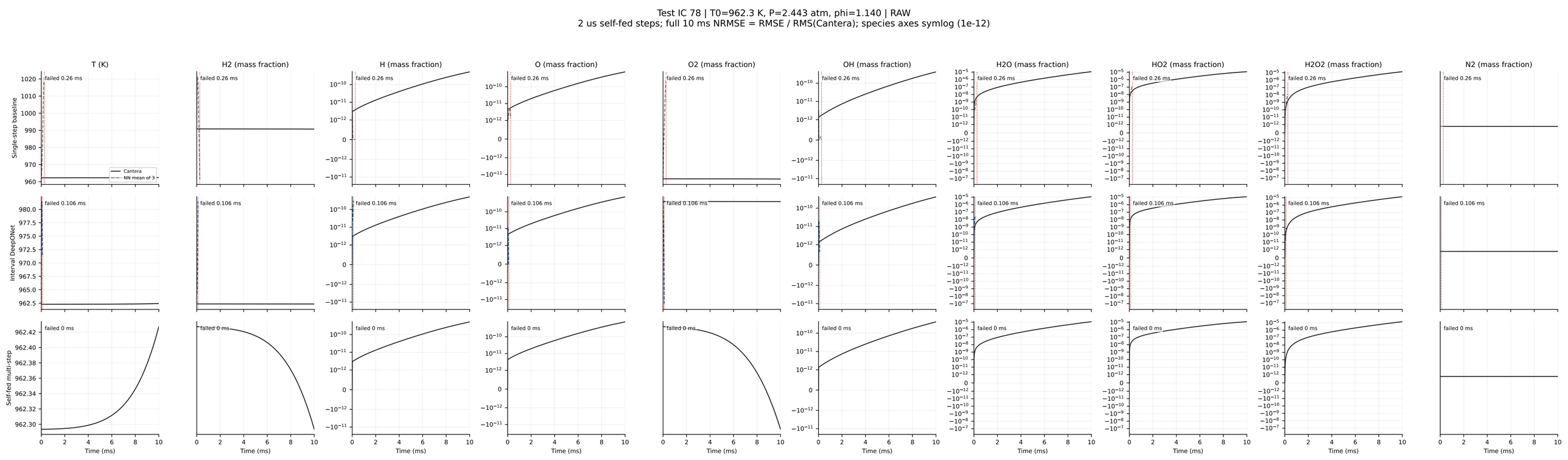

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.528 ms
failed 0.116 ms
failed 0.722 ms
Cantera
NN mean of 3
Time (ms)

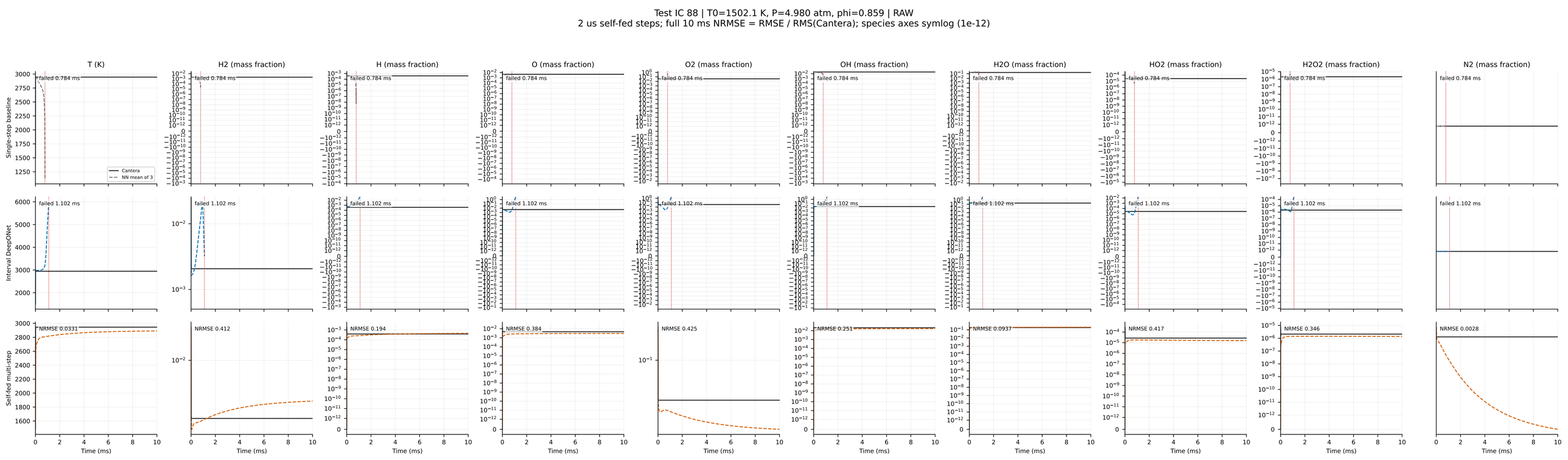

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 0.02 ms
failed 0 ms
+9.3496e2
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0487
NRMSE 0.131
NRMSE 0.381
NRMSE 0.963
NRMSE 0.214
NRMSE 0.532
NRMSE 0.0754
NRMSE 0.534
NRMSE 0.638
NRMSE 1.53e-16
failed 0.32 ms
NRMSE 0.012
NRMSE 0.0237
NRMSE 0.163
NRMSE 0.43
NRMSE 0.112
NRMSE 0.109
NRMSE 0.0398
NRMSE 0.425
NRMSE 0.518
NRMSE 0.000697
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.162
NRMSE 0.983
NRMSE 0.979
NRMSE 15.4
NRMSE 39.6
NRMSE 10.7
NRMSE 0.248
NRMSE 4.34
NRMSE 47.1
NRMSE 7.61e-16
Cantera
NN mean of 3
failed 0.658 ms
NRMSE 0.0437
NRMSE 0.19
NRMSE 0.474
NRMSE 0.566
NRMSE 0.254
NRMSE 0.314
NRMSE 0.0758
NRMSE 0.849
NRMSE 0.915
NRMSE 0.00405
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 0.276 ms
failed 1.296 ms
NRMSE 0.0765
NRMSE 0.986
NRMSE 0.413
NRMSE 0.847
NRMSE 0.824
NRMSE 0.666
NRMSE 0.0712
NRMSE 0.881
NRMSE 0.765
NRMSE 0.012
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 0.26 ms
failed 0.106 ms
failed 0 ms
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 0.784 ms
failed 1.102 ms
NRMSE 0.0331
NRMSE 0.412
NRMSE 0.194
NRMSE 0.384
NRMSE 0.425
NRMSE 0.251
NRMSE 0.0937
NRMSE 0.417
NRMSE 0.346
NRMSE 0.0028
Time (ms)